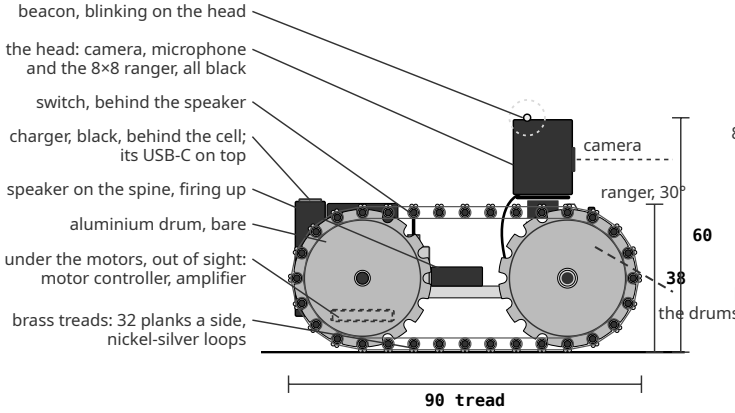
Mars rover: how the finished thing looks

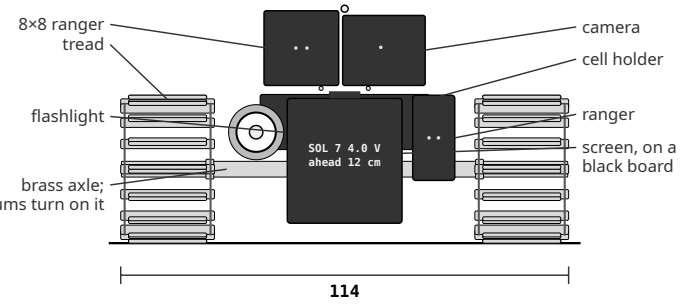
A wide little tank on a brass spine: brass treads on bare aluminium drums, six brass tubes, every part in its own colour (brass, aluminium, black), the blue and green boards out of sight. 114 × 90 × 60 mm, about 332 g; one scale.

Side view, from the rover's right

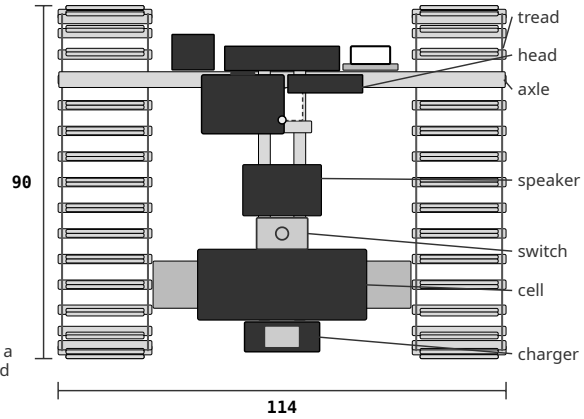


Front view

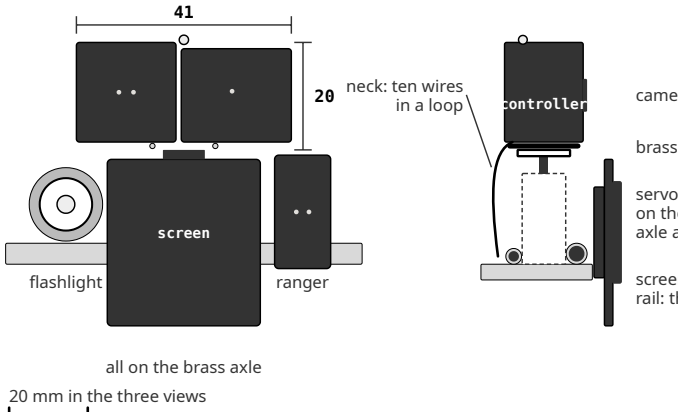
the rover's right is on your left



Top view, front up

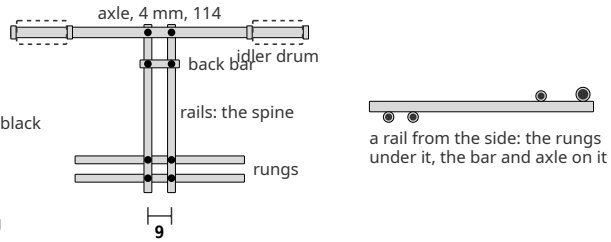


The face and a cut through the servo, at 3.6 px per mm



20 mm in the three views

The frame: six brass tubes, eight laps



a rail from the side: the rungs under it, the bar and axle on it

3 mm tube: two rails 64, two rungs 65, the back bar 15 mm
4 mm tube: the axle, 114 mm; 5 mm: four collars, 2 mm each
Six straight tubes, no bends; 8 laps soldered. The frame is the ground. The rails' open ends carry the wires inside: the bus and BAT+ in the left, the sound in the right.

Key numbers

114 × 90 × 60 mm	wide, long, tall
332 g	centre 21 mm up; tips 69°, 50° back
8.3 cm/s	on the desk; turns on the spot
6 tubes	the whole frame, 8 laps, and the ground
head 41 × 20 × 16	turns 120°; it hears and sees
64 planks	6.5 mm apart on 16-seat drums
1.9 h / 3.1 h	exploring / talking, on one 16340
2.2 h	to charge, USB-C at 500 mA
5.1 mm	clearance: the screen's edge

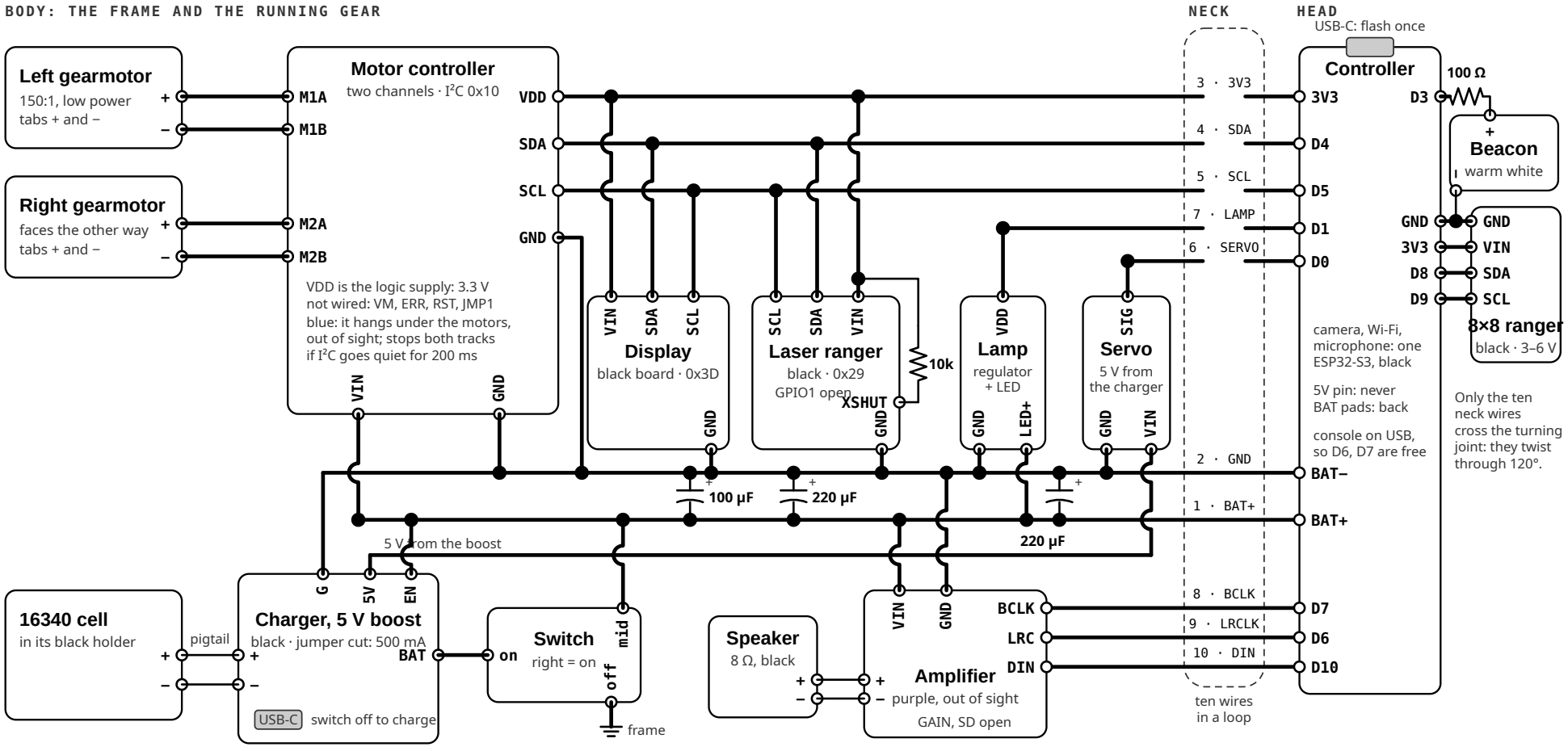
Brass, aluminium and black

Brass treads and frame, bare aluminium drums; black head, screen, ranger, charger, cell holder and speaker, as they are sold. No paint, no skins: the blue and green boards hang under the motors, out of sight. White light, one warm on the head.

Rev K for approval, 2026-11-10: brass frame, no paint, a voice. Rev J approved 2026-11-09; Rev H 2026-11-08; Rev A 2026-11-06. Nothing cut or soldered yet. Caliper the drums, the boards and the servo first.

Mars rover: every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = wire you solder (26 AWG for BAT+, 30 AWG for the rest). Thin = a lead that comes with the part. Pins are grouped by job, not by position on the board.



The budget, from the cell (mA)

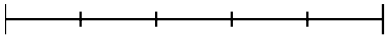
parked, watching	158 mA	Wi-Fi on, camera idle, display, ranger · 5.4 h
exploring	446 mA	camera streaming, the 8×8 ranger on, both tracks, the head sweeping · 1.9 h
at night	621 mA	exploring + the flashlight at its 50 % cap · 1.4 h
talking	278 mA	stopped, listening and speaking at a room's level · 3.1 h
never all at once	2002 mA	every stall and the lamp full; ramps and caps keep peaks near 1 A

3.3 V rail at its worst: 572 mA of the regulator's 700.
Charge: USB-C at the charger, port up, switch off, 500 mA, about 2.2 h.
Never: both USB-C ports at once · the 5V pin · the cell soldered.
Flash with the switch on, so the board's own charger sees the cell.
In the rails: BAT+, 3V3, SDA, SCL left; BCLK, LRCLK, DIN, 5V right.

Rev K for approval, 2026-11-10: a black charger with its own 5 V boost, the voice on three more neck wires, the switch grounding the supply when off. 15 crossings, each a hop. Sources: Pololu, Adafruit, Seeed, Same Sky, ST, Analog Devices.

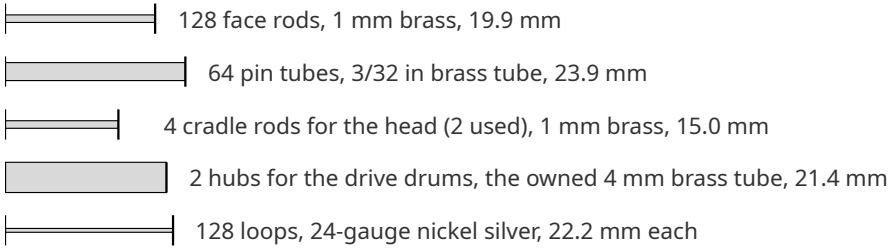
Tread and drum, actual size

Print at Actual size, then check the 50 mm bar with a ruler. Lay rods on the shapes to cut them, and build on the strip.



50 mm

1. Cut



128 face rods, 1 mm brass, 19.9 mm

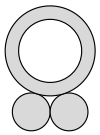
64 pin tubes, 3/32 in brass tube, 23.9 mm

4 cradle rods for the head (2 used), 1 mm brass, 15.0 mm

2 hubs for the drive drums, the owned 4 mm brass tube, 21.4 mm

128 loops, 24-gauge nickel silver, 22.2 mm each

2. Solder a plank: two rods side by side, the tube on top

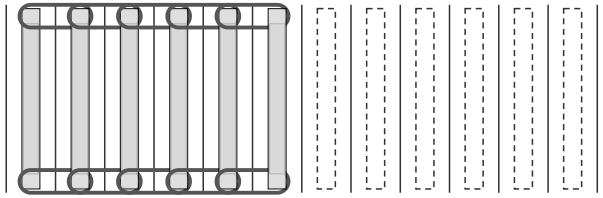


end on, 5 × size: 2.4 wide, 3.3 thick



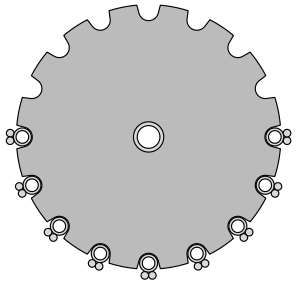
from the tread face, 2 × size: the tube stands out 2 mm each side for its two loops. Solder the rods to the tube along their length; keep the tube's ends bare.

3. Tape planks face down on the stations and close a loop round each pair of tube ends



6.5 mm stations: 32 planks close each track, the last loop closed on the rover. Each loop: 22 mm of 24-gauge nickel silver wrapped round the two pins, ends twisted, a dot of solder on the twist only.

4. Check a drum and a strip against this



Aluminium drum, two 3/8 in halves glued: 16 seats, 35.0 mm over the teeth. Each plank's pin tube sits in a round seat, 0.12 mm loose; the planks are 6.52 mm apart on the runs. Lay the strip round the drum: every pin tube should drop into a seat.

Bare aluminium, no paint: file each seat and the hole.

Drives on brass hubs; idlers turn on the 4 mm axle.

Rev K for approval, 2026-11-10. Frame: 3 mm tube, rails 64, rungs 65, back bar 15; 4 mm axle 113.5; four 5 mm collars 2. Numbers from build_page.py.

MARS ROVER · STEP 1: THE FINAL BUILD · REV K, FOR APPROVAL

A wide little desk tank on a brass spine, with a low head that looks, listens and talks

How it looks, how it is wired, every part and every pin. 114 × 90 × 60 mm, about 332 g. Rev K for approval (2026-11-10): a brass-tube frame, no paint and no skins, a voice. Rev J approved 2026-11-09. Nothing built; the drums and the Pololu parts are ordered, the rest waits on your yes.

What it is

A wide little tank that lives on a desk, built on a brass spine. Two brass treads, soldered from 1 mm brass rod and 3/32 in brass tube and joined with nickel-silver loops, run on four bare aluminium drums like the reference's. Between them, six straight brass tubes make the frame: two rails down the middle, two rungs under the motors, a short back bar, and a 4 mm axle across the front that the idler drums turn on. The servo stands on the rails; the small black screen, a black laser ranger and a flashlight hang on the axle; the cell in its black holder, a black charger, the switch and a black speaker sit on the rear deck. The head is the black camera board on its side with a black 8×8 ranger beside it and a blinking beacon on top.

You drive it from a phone and talk to it. The page shows what the camera sees; the ranger watches the desk ahead and stops the treads before an edge; the 8×8 ranger measures the room as the head turns; the camera board's microphone listens and the speaker answers. It charges over USB-C at the charger behind the cell. No box, no paint, no skins, nothing printed: every part shows its own colour (brass, aluminium, steel, black), and the blue and purple boards that have no black twin hang under the motors, out of sight.

The reference is Huy Vector's DIY Micro RC Spy Tank (video): a copper-wire tank on treads of short rods soldered side by side, with an FPV camera, two LED headlights and 16340 cells. Ours keeps the metal frame, the soldered treads, the camera, the light and the 16340, and gives it a low turning head that sees, hears and speaks.

Step 1 of the house rule. Rev K, for approval. Rev J was approved as drawn on 2026-11-09, Rev H on 2026-11-08 and Rev A on 2026-11-06. Rev K keeps Rev J's treads, drums, size and pins, and makes three changes at your word: the frame is brass tube, nothing is painted or skinned, and it has a voice. The physics page still follows Rev J; it is rewritten after the yes. Nothing cut or soldered yet.

The decisions

A BRASS SPINE, THE BOARDS ON IT

No box and no paint. Two brass treads carry a frame of six straight brass tubes, lapped and soldered: two 3 mm rails down the middle (the spine), two rungs under the motors, a short back bar, and a 4 mm axle across the front that the idler drums turn on. No bends, 8 laps. The servo stands on the rails between the axle and the back bar; the head turns on it; the screen, the laser ranger and the flashlight hang on the axle. 114 wide, 90 long, 60 mm tall, about 332 g. You drive it from a phone over Wi-Fi, watch through its camera and talk to it.

THE FRAME CARRIES THE WIRES AND THE GROUND

The rails are tubes with open ends, so the long runs go inside them: BAT+ and the I^2C bus in the left rail, the three sound lines and the servo's 5 V in the right. The frame is ground, as in a circuit sculpture: each board's GND is soldered to the nearest tube. Brass takes solder far more easily than Rev J's stainless did.

TREADS YOU SOLDER, LIKE THE REFERENCE'S

Huy Vector's treads are links of short rods soldered side by side and tied with thread round grooved drums. Ours: each plank is a 3/32 in brass tube with two 1 mm brass rods soldered under it, side by side on the tread face. The

tube is 4 mm longer; its ends are the hinge pins. A loop of 24-gauge nickel silver joins each pair of neighbouring pins. 32 planks a track, 6.5 mm apart, 128 loops in all.

LASER-CUT DRUMS, BARE

Four drums of two 3/8 in 6061 halves each, cut on SendCutSend's laser and already ordered: 19.1 mm wide, 16 round seats, 35.0 mm over the teeth. No paint: bare aluminium against the brass. The drive drums are glued on 21 mm brass hubs crimped on the gearmotors' flats. The idlers turn on the brass axle itself between two collars, so the two idler screws are gone.

NO PAINT, NO SKINS: EVERY PART IN ITS OWN COLOUR

Your rule of 2026-11-09. Every part that shows was read off its maker's photos. Black where black exists: the camera board, the screen's board, both laser rangers, the charger, the cell holder and the speaker; the drone's green 8×8 and Rev J's blue charger, green boost and green ranger give way to black ones. What only comes in another colour sits where nothing sees it: the blue motor controller and the purple amplifier hang under the motors, the translucent-blue servo stands under the head and behind the screen, and the switch's blue body stands between the holder and the speaker, behind the screen. The cell's yellow print faces down. Lights stay white: the charger's status lights face the holder, and the ranger's power light comes off.

BRASS, ALUMINIUM AND BLACK

Brass treads and frame; bare aluminium drums; steel motor cans; black head, screen, rangers, charger, holder and speaker. White text on the screen, a white flashlight, one warm beacon. No green, blue, red or purple in sight, no card, no sheet, no paint, nothing printed. Prototype, not product: the head pulls off its horn, the firmware updates over the air.

A CAMERA, A MICROPHONE AND AN 8×8 RANGER IN THE HEAD

The controller board is the head: one 21 × 17.8 × 15 mm stack with the camera and a digital microphone, its boards black. Pimoroni's black 8×8 laser ranger stands beside it: two eyes on a black face 41 mm wide. Both sit on a cradle of two 1 mm brass rods on the servo's horn, lifted 1.5 mm so the head turns clear of the taller screen. Ten silicone wires join the head to the frame in a loose loop: they twist rather than bend through its 120°.

A VOICE: THE OWNED SPEAKER AND AMPLIFIER

The microphone is on the camera board already. The voice is a spare enclosed speaker, black, lying on the spine behind the head and firing up, driven by a spare I^2S amplifier hung under the motors. The controller sends it sound on three pins (D6, D7, D10), down three more neck wires and through the right rail. It draws about 120 mA while it speaks, so the treads wait while it talks (and while it listens, because their patter fills the microphone). Nothing to buy.

A MOTOR CONTROLLER ON THE SAME TWO WIRES

The motor controller runs motors from 1.8 V, so one cell drives it directly. It takes speed commands over I^2C, stops both tracks if the commands stop (a 200 ms timeout), ramps acceleration, and reports the cell's voltage. Its board is blue, so it hangs under the motors.

THE HEAD TURNS ON A SERVO ON THE SPINE

The servo stands shaft up on the two rails, glued between the axle in front and the back bar behind. The head's cradle sits on its horn: 120° on the datasheet's pulses, about 180° on wider ones. It is rated 4.8 to 6 V, so it runs on the black charger's 5 V boost, which the switch turns off with everything else.

THE LASER RANGER IS THE ROVER'S HAZARD EYE

The black ranger hangs on a brass stub on the axle, left of the screen, tilted 30° down. Its centre ray meets the desk 4.6 cm in front of the rover, 5.4 cm away: past its 4 cm minimum, about half a second ahead at full speed.

When the desk disappears ahead, the tracks stop. It is the only sensor that sees a drop, so reversing is limited to 2 cm.

AN 8×8 RANGER IN THE HEAD, FOR THE ROOM

It reads 64 distances at once over a 45° square, 2 cm to 4 m. The head's 120° turn sweeps it, so one sweep sees about 165° ahead: walls, chair legs, the far side of the desk. The frames go to the desktop with the pictures, and the desktop builds a rough map of the room. It draws about 95 mA while it ranges and sleeps when the rover is parked.

THE DRONE'S FLASHLIGHT, AS A HEADLIGHT

The same cool-white LED on a star sawn to 14 mm, the same 10 mm lens, and one 350 mA regulator whose supply pin is the controller's pin D1: on, off and dimming with no MOSFET. Its tab is soldered to the brass axle, a better heatsink than Rev J's heat-shrink. About 140 lm in a 23° spot, right of the screen, pointing straight ahead; capped at 50 % in firmware.

A BEACON THAT BLINKS ON THE ROOF

The satellite's pre-wired warm-white chip, glued flat on the head's roof, 100 ms every 2 s through 100 Ω. It says the rover is on, and it turns with the head.

ONE CELL, WHERE THE RTG GOES

The owned 16340 cell in its black holder lies across the rear deck on the two motor cans. Behind it stands the black charger, port up; in front, the owned toggle, lever to the right for on. Plug in with the switch off: about 2.2 h to full. Off grounds the rover's supply, which also turns off the charger's 5 V boost, so nothing drains the cell. About 1.9 h exploring with the camera streaming, 1.4 h with the flashlight on, 3.1 h stopped and talking, 5.4 h parked and watching.

A REFERENCE, NOT A COPY

Huy Vector's spy tank is a copper-wire box on hand-soldered metal treads, with three cells, a toy RC receiver, an analog FPV camera and two LED headlights on a resistor. This one keeps the idea: a palm-sized tracked chassis on a metal frame, treads of soldered rods, a camera, a light, a 16340 and USB-C charging. Everything else is its own: a wide little tank on a brass spine with a low turning head that sees, hears and speaks, laser-cut drums of two halves, metal loops instead of thread, Wi-Fi to a phone, a laser ranger that keeps it on the desk, one cell instead of three, brass and black instead of copper.

Dimensions

WHAT	MM
overall width, pin end to pin end	113.9
overall length	89.8
overall height, desk to the beacon on the roof	59.6
tread height, round a drum	37.6
plank length (the tread face)	19.9
plank pitch	6.5
drum centres	52.1
the tread's hinge line round a drum (radius)	16.7

drum, over the teeth	35.0
drum, wide (two 3/8 in halves)	19.1
each drive drum's brass hub	21.4
drum's inner face, from the centre line	35.5
the front axle's ends, from the centre line	56.8
tread centre line, from the rover's centre line	45.0
gearbox face, from the centre line	32.8
motors' inner ends, from the centre line	7.2
the two rails (the spine), from the centre line	4.5
rails above the desk	15.3
rungs above the desk (the motors sit on them)	12.3
front axle above the desk (the drums' height)	18.8
back bar, ahead of the body's centre	14.1
back bar above the desk	18.3
the face (screen, ranger, lamp), ahead of the body's centre	34.6
screen top edge	36.8
screen bottom edge, its tabs	5.1
servo shaft, ahead of the body's centre	19.8
servo's foot, on the rails	16.8
servo's top face	34.1
head's floor above the desk	38.6
camera lens, ahead of the body's centre	27.9
camera lens above the desk	49.0
ranger above the desk	26.8
where the ranger's centre ray meets the desk, ahead of the face	46.4
the ranger's distance to that point	53.6
cell holder's floor (on the motor cans)	23.8
speaker's back edge, on the spine	-8.6
underside of the boards under the motors	7.8
ground clearance: the lowest part, the screen's tabs	5.1

MASS

PART	G
brass treads: 64 planks and 128 nickel-silver loops	54.6
four aluminium drums of two halves each, the two drive hubs, four collars	181.3
two gearmotors	19.0
cell and holder	26.0
charger with its boost, switch	5.5
motor controller and amplifier, under the motors; the 100 and 220 µF	4.5
speaker, on the spine	1.6
servo, on the rails	4.4
laser ranger, lamp and its regulator, the 220 µF	4.0
display	3.0
frame: six brass tubes	14.4
head: controller with camera and microphone, the 8x8 ranger, beacon, cradle	9.1
wire, Kapton, glue	4.5
Total	**332**

Centre of mass 21 mm up and 3 mm behind the middle. It tips sideways at 69° and backwards at 50°. The treads', drums' and head's masses are computed or estimated; weigh them.

Pin map, XIAO ESP32-S3 Sense (in the head)

The I²C pair, the three sound pins and two signals leave the head down the neck; everything else in the body is on I²C.

PIN	GPIO	JOB	GOES TO
D0	1	servo signal, 50 Hz, 900 to 2100 µs for 120°	neck wire 6, down to the servo standing on the rails
D1	2	flashlight: brightness by PWM, capped at 50 %	neck wire 7, down to the regulator's VDD pin behind the lamp (it draws under 1 mA)
D2	3	free	nothing: a strapping pin (JTAG source), left alone
D3	4	beacon, high = lit	100 Ω, then the beacon chip's + lead, on the head's roof
D4	5	I²C SDA, 400 kHz	neck wire 4, down to the ranger, the display and the motor controller
D5	6	I²C SCL	neck wire 5, down to the same three

D6	43	I²S word clock (LRCLK), 16 kHz	neck wire 9, down to the amplifier's LRC; the console moves to the USB port, so UART0 is free
D7	44	I²S bit clock (BCLK), 512 kHz	neck wire 8, down to the amplifier's BCLK
D8	7	I²C SDA of a second bus, 1 MHz	the 8×8 ranger's SDA, in the head
D9	8	I²C SCL of the second bus	the 8×8 ranger's SCL, in the head
D10	9	I²S data out (DIN), 16-bit mono	neck wire 10, down to the amplifier's DIN; the Sense board's SD slot shares D8 to D10, so the slot stays empty
mic (inside)	42, 41	the Sense board's digital microphone, PDM clock and data	nothing to wire: it is on the camera board, its port facing forward
3V3		3.3 V out of the board's regulator, 700 mA available	neck wire 3, down to the ranger VIN, the display VIN and the motor controller VDD; in the head, the 8×8 ranger's supply pin
GND		ground	the beacon and the 8×8 ranger, from the header's GND pin; neck wire 2 at the BAT− pad
BAT+ / BAT− pads		the cell, up the neck, after the switch	two pads on the back of the board; neck wires 1 and 2
5V		USB VBUS only: no voltage on battery	nothing, ever
USB-C		flashing, once, on the bench; updates over Wi-Fi after that. It stays open on the head's left end	a computer; flash with the switch on

THE I²C BUS

laser ranger `0x29` · display `0x3D or 0x3C, by the jumper on its back` · motor controller `0x10, plus the general call 0x00` · 8×8 ranger, on its own bus (D8, D9) `0x29, the laser ranger's address too: hence the second bus`. No two share an address, so nothing is re-addressed at boot.

MOTOR CONTROLLER PINS

Used: `VIN` and the `GND` beside it (the cell, after the switch), `M1A` `M1B` (left motor), `M2A` `M2B` (right motor), `VDD` (3.3 V from the head), `SDA`, `SCL` and one logic `GND`. Not wired: `VM`, `ERR`, `RST`, `JMP1`.

Every wire, and why

NET	FROM	TO	WIRE	WHY
CELL	the holder's two leads	JST-PH pigtail into the charger's socket	the holder's leads soldered to the owned pigtail, in heat-shrink, behind the charger	The cell never meets an iron: the joint sits 40 mm from it. The button top is +. Meter the plug with the cell in before it goes into the socket.

VBAT	charger BAT pad	switch, the end lug the middle meets with the lever right (meter which)	26 AWG, 20 mm	The charger's BAT pad is the cell itself. Lever right joins it to the rover; lever left leaves the charger alone with the cell, so it charges cleanly.
BAT+	switch, middle lug	motor controller VIN · the 100 µF · amplifier VIN · its 220 µF · the charger's EN pad · through the left rail: the LED + pad · the 220 µF · neck wire 1	26 AWG	Everything runs straight off the cell, 3.0 to 4.2 V. The motor controller works from 1.8 V, the amplifier from 2.5 V, the controller board has its own regulator, and the flashlight's regulator is a current sink. The charger's EN pad rides on this rail, so its 5 V boost is on only while the rover is.
OFF	switch, the other end lug	the frame (ground)	a 5 mm stub of 1 mm brass rod to the rail under it	Lever left ties BAT+ to ground: the capacitors empty, the charger's EN pad goes low and its boost stops, so nothing drains the cell while the rover is off. A toggle breaks before it makes, so the cell never meets ground.
GND	charger G pad, on the rear rung	the frame · motor controller GND (power side) · the 100 µF · amplifier GND · the 220 µFs · servo − · regulator tab · neck wire 2	the frame, and 26 AWG where a board sits off it	The frame is ground, as in a circuit sculpture: each board's GND is soldered to the nearest tube. The motors' current returns through the rear rung, a few millimetres of brass from the charger's G pad; the head's comes down the neck on its own wire.
5V	charger 5V pad	through the right rail to the servo + lead	30 AWG, then the servo's own lead	The servo is rated 4.8 to 6 V, so it does not run on the cell. The charger's boost makes 5 V at up to 1 A, more than the servo's 0.4 A stall. It will not start under more than 200 mA, so the firmware waits for boot before it moves the head.
3V3	neck wire 3	display VIN · ranger VIN, and its XSHUT through 10 kΩ · through the left rail: motor controller VDD	30 AWG	The body's logic runs on the head's 3.3 V regulator. That matters for the motor controller: its 10 kΩ I^2C pull-ups go to VDD, and VDD must not be the cell, or the bus would sit at 4.2 V on pins rated 3.6. The ranger module and the display take 3 to 5 V and shift their own levels.
SDA, SCL	neck wires 4 and 5	display · ranger · through the left rail: motor controller	30 AWG	One I^2C bus for the body: the ranger and the display at the front, the motor controller at the back, the long run inside the left rail with BAT+ and 3V3 and away from the sound lines in the right. The boards' pull-ups in parallel come to a few kΩ: fine at 400 kHz. The ranger's GPIO1 stays unconnected.
GND (logic)	display GND	ranger GND · motor controller logic GND	the frame	The display's and the ranger's GND pins are soldered to the axle and the brass stub; the motor controller's to the rear rung.
M1, M2	motor controller M1A, M1B, M2A, M2B	left and right gearmotor tabs	30 AWG, 30 mm, under the motors	Each motor's two tabs. The right motor faces the other way, so the firmware flips its sign. The stall current at 3.7 V is 0.22 A, so 30 AWG is ample. The cans sit on the grounded rungs: meter each can to its tabs, open.

SERVO	neck wire 6	servo signal lead	30 AWG, then the servo's own lead, cut to 30 mm	The servo takes a 3.3 V signal (FEETECH, via Pololu). Its three leads stay under the head.
LAMP	neck wire 7	regulator VDD	30 AWG	The controller's pin is both the switch and the dimmer: the regulator wakes above 2.7 V on VDD and draws under 1 mA there, the drone's trick.
LED−	LED − pad on the star	regulator OUT	the LED's own − lead	The regulator sinks 350 mA from the LED's cathode to ground, whatever the cell's voltage, while the cell stays 0.12 V above the LED's 2.9 V. Its tab is soldered to the axle, its heatsink.
BAT+ (head)	neck wire 1	controller BAT+ pad	26 AWG in the neck	Ten silicone wires in a loose loop beside the servo's horn: they twist, not bend, through the head's 120°.
GND (head)	neck wire 2	controller BAT− pad; from its header GND pin, the beacon − and the 8×8 ranger's GND	26 AWG in the neck, 30 AWG in the head	One ground in the head, joined at the controller.
3V3, SDA, SCL (head)	controller 3V3, D4, D5	neck wires 3, 4, 5	30 AWG	The bus leaves the head on three wires; nothing on it stays up there.
BEACON	controller D3	100 Ω, then the beacon chip's + lead	the chip's own enamelled leads	A warm-white 0805 chip drops about 2.9 V: 100 Ω from 3.3 V gives 4 mA, the satellite's figure.
3V3 (8×8)	controller 3V3 pin	8×8 ranger's supply pin	30 AWG, 15 mm	Pimoroni's board takes 3 to 6 V on one pin and keeps its sensor and its I²C pull-ups at 3.3 V, so the controller's 3.3 V feeds it directly. About 95 mA while it ranges, 150 at worst; the regulator gives 700.
SDA2, SCL2	controller D8, D9	8×8 ranger SDA, SCL	30 AWG, 15 mm	A second I²C bus, all inside the head: the 8×8 ranger answers at 0x29, the laser ranger's address, so it gets its own two pins instead of an address change. Its INT pin stays unconnected.
BCLK, LRCLK, DIN	controller D7, D6, D10	amplifier BCLK, LRC, DIN	30 AWG: neck wires 8, 9, 10, then through the right rail	The voice. The controller sends 16-bit mono sound at 16 kHz as I²S: a 512 kHz bit clock, a 16 kHz word clock and the data. The three run down the neck and through the right rail to the amplifier under the motors, away from the I²C pair in the left rail. The firmware moves its console to the USB port, so D6 and D7 (UART0) are free; at reset the ROM still prints on D6 for a moment, and the amplifier stays quiet because it sees no bit clock.
SPK+, SPK−	amplifier + and −	speaker leads	the speaker's own 32 AWG leads, under it	A bridged output, 8 Ω: neither lead is ground, so neither touches the frame. About 120 mA from the cell while it speaks at a room's level, 0.4 A on a loud syllable, which the 220 µF at the amplifier covers. GAIN stays open (9 dB) and SD on its board's pull-up, so it plays the mono mix.

The budget

LOAD	MA	SOURCE
controller, Wi-Fi on	110	Seeed wiki, at 3.8 V on BAT, with the Sense board
controller, camera streaming	155	the wiki's webcam figure; 366 mA at an image capture
laser ranger	20	Pololu, typical; 40 mA peaks
display	12	about half the pixels lit
motor controller logic	4	estimate
servo at rest, through the charger's boost	8	5 mA at 5 V (datasheet); the boost's 1 mA on top
8×8 ranger, ranging	95	ST's figure for the sensor; asleep when parked
amplifier idle; speaking	2	datasheet; about 120 mA while it speaks at a room's level, 400 on a loud syllable
servo moving	79	50 mA at 5 V
two motors on the desk	120	estimate; a stall is 222 mA each
flashlight at its cap	175	350 mA at 50 %

Parked and watching: 158 mA, 5.4 h on the 950 mAh cell (90 % usable). Exploring with the camera streaming and the head sweeping: 446 mA, 1.9 h. With the flashlight on: 621 mA, 1.4 h. Stopped and talking: 278 mA, 3.1 h. Charging: 2.2 h at 500 mA, switch off.

Everything at once, every stall and the lamp full, would be 2002 mA, near the cell's 2 A cut-out. It never happens: the motor controller ramps, the firmware caps the drive at 70 % and the lamp at 50 %, the head waits while the tracks start, and the treads wait while it talks. Real peaks sit near 1 A. The head's 3.3 V regulator sees 572 mA at worst, of 700.

Speed: 90 rpm at 6 V is 56 rpm at 3.7 V; the tread moves with its pins, 33 mm across a drum, so 97 mm/s free and about 83 on the desk.

Parts

Rev K adds four black parts: one on Amazon, two from DigiKey in one order, one from SB Components in the UK (no US shop had it). It drops two lines from your Amazon cart, the brass sheet and the M3 screws. The drums and the Pololu parts are ordered. Owned parts are listed so they are not bought twice, and Rev J parts that Rev K leaves out are listed for the shared bin. Each part is named by what it is, with the link on the word to the exact product.

NEW FOR REV K: FOUR LINES, EACH BLACK AS SOLD

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

1	USB-C charger with a 5 V boost, black	Adafruit bq25185 board; DigiKey 1528-6106-ND	$8.95	One black board does two jobs: it charges the cell over USB-C and makes the servo's 5 V. **Cut the jumper on its back for 500 mA**: a full cell in about 2.2 h. It stands behind the cell holder, port up, with its parts and its three status lights facing the holder, so only its black back and the USB-C's metal show. If its green terminal block comes soldered, lift it off: the 5 V comes off the pad under it. It replaces Rev J's blue charger and green boost.
1	0.96 inch OLED display, white, I²C, on a black board	Adafruit; DigiKey 1528-1220-ND	$17.50	The face. The five owned screens sit on blue boards; this one is black with gold pads. 29.2 × 26.7 mm, with two tabs below (31.8 in all); the glass is 26.6 × 19.3. It is glued to the axle by its back. Adafruit's current photos show the black board: if a blue one arrives, send it back.
1 (a 2-pack)	laser ranger, 4 m, black, with a cover	AITRIP VL53L1X module; Amazon	$17.99 for two	The desk-edge eye: 25 × 10.7 × 3.2 mm (listing), 3 to 5 V, its own level shifting. Its GND pin is soldered to a 1 mm brass stub on the axle, left of the screen, tilted 30° down. **XSHUT goes to its VIN through 10 kΩ**; GPIO1 stays open. Check it answers at 0x29 on the bench. If it has a power light, lift the light off. The second module goes to the shared bin.
1	8×8 laser ranger, 4 m, 63°, black	Pimoroni VL53L5CX breakout; SB Components (UK), in stock 2026-11-10	$33.00 + shipping	The room eye. It is black, with its parts on the back so the face is flat: 19 × 19 × 4.6 mm, 3 to 6 V, address 0x29, 2 cm to 4 m, about 95 mA while ranging. Pimoroni, DigiKey and The Pi Hut were sold out on 2026-11-10. The drone's green 8×8 stays the drone's spare.

IN YOUR AMAZON CART FROM REV J: FOUR LINES TO KEEP, TWO TO DELETE

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	controller with camera and microphone, ESP32-S3	Seeed XIAO ESP32-S3 Sense; in the cart	$25.99	The head, black: camera, Wi-Fi, a digital microphone and the brain in one 21 × 17.8 × 15 mm stack. It lies on its side on a brass cradle on the servo's horn, camera forward, its USB-C open on the head's left end. Flash it once on the bench; updates come over Wi-Fi. Its stick-on antenna goes flat on the stack's back. Keep its SD slot empty.
1 pack	1.0 mm round brass rod, 300 mm, 5 rods	K&S; in the cart	$9.03	The planks' face rods, the head's cradle and the stubs that hold the ranger and the lamp: 2.6 m, about 10 rods with the offcuts. Fifteen are owned and about eight spare, so one pack makes enough.
2 packs of 3	round brass tube, 3/32 in, 0.014 in wall, 12 in	K&S; in the cart	$8.05 a pack	The planks' pins: 64 pieces of 23.9 mm, 12 to a tube. Each sits in a drum's seat 0.12 mm loose; the loops wrap its ends.
1 spool	24-gauge nickel-silver wire, dead soft, 25 ft	CRAFT WIRE; in the cart	$24.99	The tread's 128 loops, 2.8 m. The owned 20 gauge would hang below the tread's face and drag; 24 gauge stays a quarter of a millimetre clear.
delete	brass sheet, 0.005 in, 6-pack	K&S; in the cart	$24.81	**Delete it before checkout.** It was for Rev J's ivory skins, which the no-paint, no-cover rule took away.

delete	M3 × 25 hex-socket screws, 20	uxcell; in the cart	$7.39	**Delete it before checkout.** The idler drums now turn on the brass axle, so they need no screws.

ORDERED FOR REV J AND STILL IN THE BUILD

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
10	aluminium drum half, laser cut: 16 seats, 35 mm, 3/8 in 6061	SendCutSend order SJ983673, ordered 2026-11-09	$42.10 for ten	The four wheels, two halves each, glued face to face: 19.1 mm wide, 35.0 mm over the teeth, bare aluminium all over now (no paint). The two drive drums are glued on brass hubs on the motors' shafts. The two idlers have no hub: they turn on the 4 mm axle itself, their 4.1 mm holes 0.05 mm loose, between two brass collars. Run a round needle file through each seat and each hole; two halves are spares.
2	micro metal gearmotor, 150:1, low power, 6 V	Pololu; DigiKey order 102785005, shipped 2026-11-09	$23.95 each	The drive. About 56 rpm on the cell: 8.3 cm/s on the desk. Steel cans and gearboxes, their own colour. They sit on the two rungs, the cell holder on top of them.
1	motor controller, two channels, I^2C	Pololu; DigiKey order 102785005	$15.95	Its board is blue, and no black two-channel controller this small takes I^2C. So it hangs under the motors, parts down, glued to the rungs with its GND soldered to one: out of sight. It runs the motors from 1.8 V, stops them if the controller goes quiet, and reads the cell.
1	sub-micro servo, 120°	Pololu; DigiKey order 102785005	$10.21	Its case is translucent blue, so it stands where nothing sees it: on the two rails, between the axle and the back bar, under the head and behind the screen. Its white horn carries the head. It runs on the charger's 5 V.

ORDERED FOR REV J, LEFT OUT OF REV K: TO THE SHARED BIN

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	USB-C Li-ion charger, 500 mA (blue)	Adafruit; DigiKey order 102785006	$5.95	Replaced by the black charger with its own boost. Unopened, DigiKey may take it back.
1	5 V boost regulator (green)	Pololu; DigiKey order 102785005	$8.95	The black charger makes the servo's 5 V now.
1	laser ranger, 4 m (green)	Pololu; DigiKey order 102785005	$22.95	Replaced by the black module. Crumb's edge ranger is the same green board, so it may go there.

OWNED: POWER, SWITCH AND SOUND

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1 of 13	16340 cell, 950 mAh, protected, button top	Nitecore NL169, IMR Batteries 2026-10-16	owned	16.6 × 34.1 mm, 20 g, 2 A protection. Its wrap is black with yellow bands and print: it lies in the holder with its print face down, like a rover's RTG across the rear deck.

1 of 10	16340 holder with leads	ACEIRMC, Amazon 2026-10-27	owned	43 × 18 × 14 mm, black, glued across the two motor cans. Its leads run down its back, behind the charger, to a JST-PH pigtail.
1 of 20 pairs	JST-PH 2.0 mm pigtail	daier set, Amazon 2026-09-02	owned	Into the charger's socket, between the charger and the holder, out of sight. **Meter its polarity, not its colours.**
1 of 20	toggle switch, on-on	Ferwooh MTS-102, Amazon 2026-10-27	owned	Its body is blue and no black one turned up (the black-looking candidate at DigiKey is red), so it stands on the two rails across the rover, its back to the holder and the speaker in front of it: the screen hides it from the front, the holder from behind, the rear drums from the sides; its chrome top and lever show. Lever to the rover's right is on. Wired so off grounds the rover's supply.
1 of 6	enclosed speaker, 8 Ω, 0.8 W	Same Sky, DigiKey 2026-09-02	owned	The voice. Black, 20 × 13 × 4.9 mm, a sealed back, firing from its face. It lies on the two rails behind the head, firing up; its leads run under it to the amplifier.
1 of 12	I^2S class-D amplifier, 3 W	HiLetgo MAX98357A, Amazon	owned	Purple, 19 × 16 mm, so it hangs under the motors beside the motor controller, parts down, out of sight. VIN from the cell after the switch, with a 220 µF beside it; GAIN open (9 dB); SD on its own pull-up, so it plays the mono mix.

OWNED: FACE AND HEAD

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1 of 4	cool-white LED on a 20 mm star	Cree XP-G2 Indus Star, the drone's pack	owned	Sawn to a 14 mm disc as the drone's (solder the leads first). Bare aluminium. 139 lm at 350 mA.
1 of 2	10 mm narrow-spot lens	Carclo, the drone's DigiKey order	owned	Clear, glued over the LED; the star is glued to a 1 mm brass bracket on the axle, right of the screen, so the lens comes flush with the glass.
1 of 10	350 mA LED regulator	AMC7135, the drone's pack (arriving 2026-11-17 to 11-20)	owned	Black, behind the star: its tab (GND) soldered straight to the axle, which is its heatsink, OUT on the LED's − lead, VDD on the LAMP wire. Read the datasheet's top view before soldering.
1 of 20	pre-wired warm-white LED chip, 0805	DiCUNO 20-pack, the satellite's	owned	The beacon, glued flat on the camera board's top edge.

OWNED: FRAME, TREADS AND BENCH STOCK

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
272 mm	brass tube, 3 mm, 0.5 mm wall	M METERXITY 2 to 7 mm sets (three 200 mm sets on hand, more arriving)	owned	The frame: two rails 64 mm, two rungs 65 mm, the back bar 15 mm. Cut with the jeweller's saw, ends filed square. 8 laps, soldered with the Harris flux and rinsed.
156 mm	brass tube, 4 mm, 0.5 mm wall	the same sets	owned	The front axle, 113.5 mm, and the two drive hubs, 21.4 mm each (crimped on the motors' flats over 9 mm of shaft).

`8 mm`	brass tube, 5 mm, 0.5 mm wall	the same sets	owned	Four collars, 2 mm each, on the axle either side of each idler drum: the inner ones soldered before the drum goes on, the outer ones glued after.
`1 + 1 + 2 + 1`	100 Ω; 100 µF; 220 µF; 10 kΩ	LuminologyPro resistor kit, ALLECIN electrolytic kit	owned	100 Ω for the beacon; 100 µF at the motor controller's VIN; 220 µF at the lamp and the servo, and one at the amplifier; 10 kΩ on the ranger's XSHUT.
`about 0.8 m`	26 and 30 AWG silicone wire, black	TUOFENG 26 AWG, CBAZY 30 AWG	owned	26 AWG for BAT+, 30 AWG for the rest. Black only: most runs go inside the two rails, the rest under parts.
	Kapton tape, heat-shrink, solder, Harris flux, baking soda, alcohol	bench stock	owned	Kapton and heat-shrink as insulation only (joints, and wherever a wire crosses a tube), never as a colour cover. 63/37 and the Harris flux for the treads and the frame, rinsed with baking soda and alcohol before any board goes on.

What the firmware needs

Not written yet; listed so the pin map above is known to be enough.

- ESP-IDF 6.0.2 for the esp32s3, the family's toolchain. The `esp32-camera` component streams JPEG to a page served by the controller; the phone's browser shows the picture, a drive pad, a head slider, the lamp, and a Scan button.
- I²C at 400 kHz: ST's driver for the ranger (short mode, 50 Hz), the display through `esp_lcd`'s driver for its chip, and the motor controller's protocol from Pololu's user's guide: command timeout 200 ms, a gentle acceleration limit, speeds as signed numbers, the cell's voltage read once a second.
- A second I²C bus on D8 and D9 at 1 MHz for the 8×8 ranger: ST's ULD driver loads its firmware at boot, then 8 × 8 zones at 15 Hz while exploring, asleep when parked. Each frame carries the head's angle, so a sweep from −60° to +60° sees about 165° ahead; the frames stream to the desktop with the pictures, and the desktop builds a rough map of the room and sends back where to go.
- LEDC: the servo on D0 at 50 Hz (900 to 2100 µs, the head's 120°), down neck wire 6; the lamp on D1, a few kHz, set on the bench so the picture shows no banding, duty capped at 50 %; the beacon on D3, 100 ms every 2 s.
- Safety, in this order: the ranger reading the desk ahead every 20 ms while the tracks move forward, stopping and backing 2 cm when the reading jumps past the desk's distance; the 8×8 ranger's middle rows stopping it for anything within 10 cm ahead; reverse limited to 2 cm; the motor controller's own timeout; the cell below 3.4 V parks everything but the beacon.
- Sound, Rev K: the Sense board's microphone on I²S0 as PDM (clock GPIO42, data GPIO41), 16 kHz; the amplifier on I²S1, 16 kHz 16-bit mono out on D7 (bit clock), D6 (word clock) and D10 (data). The console moves to the USB Serial/JTAG port so UART0's two pins are free. The treads stop while it listens (their patter fills the microphone) and while it speaks (the current). The desktop that already runs the Pocket AI's server can be its voice: microphone frames up the Wi-Fi, speech back down.
- The screen: the distance ahead as a bar, the cell's percentage, the Wi-Fi name and address, and a line of text from the phone.
- Over-the-air updates after the first flash on the bench. The servo's 5 V comes on with the switch; the firmware waits half a second after boot before it moves the head, because the charger's boost will not start under a heavy load.

Before power, with the meter

- [ ] Polarity, once: the holder's pigtail plug against the charger's socket with the cell in, button top to the pin the board marks +, before the plug goes in.
- [ ] Supply, once: switch on, the head not yet connected, the cell's voltage between neck wires 1 and 2 at their free ends, and 5.0 V on the charger's 5V pad before the servo's red lead goes on it; switch off, both read 0.
- [ ] The frame is ground: with the cell out, the frame to GND reads 0 Ω and the frame to BAT+ is open. Nothing but grounds touches a tube bare: BAT+ and every signal stay in silicone, inside the rails and out, and each rail's two open ends are filed smooth before the wires go in.
- [ ] Isolation, once: each motor's can to each of its two tabs reads open, because the cans sit on the grounded rungs; and each speaker lead to the frame reads open, because the amplifier's outputs are bridged.

Rev K, for approval

What changed from Rev J, and why.

- **A brass frame.** You asked for brass tubes. Tube cannot be rolled into Rev J's idler eyes, so the frame is six straight tubes, lapped and soldered: a spine of two rails between the motors, two rungs, a back bar, and a 4 mm axle that the idler drums turn on. The two idler screws and their washers are gone. Brass takes solder easily, and the rails carry the long wires inside.
- **No paint, no skins.** Your rule of 2026-11-09. Rev J's ivory skins and painted drum faces are gone; the drums are bare aluminium. Every part that shows is black as sold: the screen's board, both laser rangers and the charger are new black parts. The blue motor controller, the purple amplifier and the translucent-blue servo sit where nothing sees them. Rev J's blue charger, green boost and green ranger go to the shared bin; the drone's green 8×8 stays the drone's.
- **A voice.** The camera board's own microphone, a spare speaker and a spare amplifier: three more neck wires, ten in all, and nothing more to buy. The treads wait while it talks.
- **The numbers.** 114 × 90 × 60 mm (Rev J 115 × 90 × 58), about 332 g (Rev J 328). The new screen is taller, so the head rides 1.5 mm higher; the ground clearance is 5.1 mm, under the screen's two tabs. 1.9 h exploring, 3.1 h talking, 2.2 h to charge.
- **The carts.** Delete the brass sheet and the M3 screws from your Amazon cart (five lines left, $76.11) and add the black ranger 2-pack ($17.99). The black charger and screen are one new DigiKey order ($26.45 before shipping). The black 8×8 comes from SB Components in the UK ($33 plus shipping): on 2026-11-10 it was the only shop with stock.

Earlier revisions

Rev A, approved 2026-11-06: silicone tracks on a stainless ladder. Rev H, approved 2026-11-08: soldered brass treads on laser-cut aluminium drums like the reference's, an open stainless frame, ivory skins. Rev J, approved 2026-11-09: Rev H on 35 mm drums, with the drone's 8×8 ranger in the head in place of Rev I's LiDAR, which was too big. Revs B to G and I were set aside.

Built by `docs/pages/gen/build_page.py`, which also writes `docs/FINAL_BUILD.md` and the PDFs; the dimensions, pins, nets, budget and parts live in that script. The wiring sheet's clearance check passes, and the drum, plank, frame, servo and head clearances are asserted. Sources: the reference page and its video, Pololu's pages and guides, the Seeed XIAO ESP32-S3 wiki, Adafruit's pages and guides for the charger with its boost (schematic read 2026-11-10) and the screen, the ranger module's Amazon listing, Pimoroni's 8×8 breakout, Same Sky's speaker, the MAX98357A and AMC7135 datasheets, and the family's inventories and records. Colours read off each maker's photos 2026-11-10. Nothing measured.